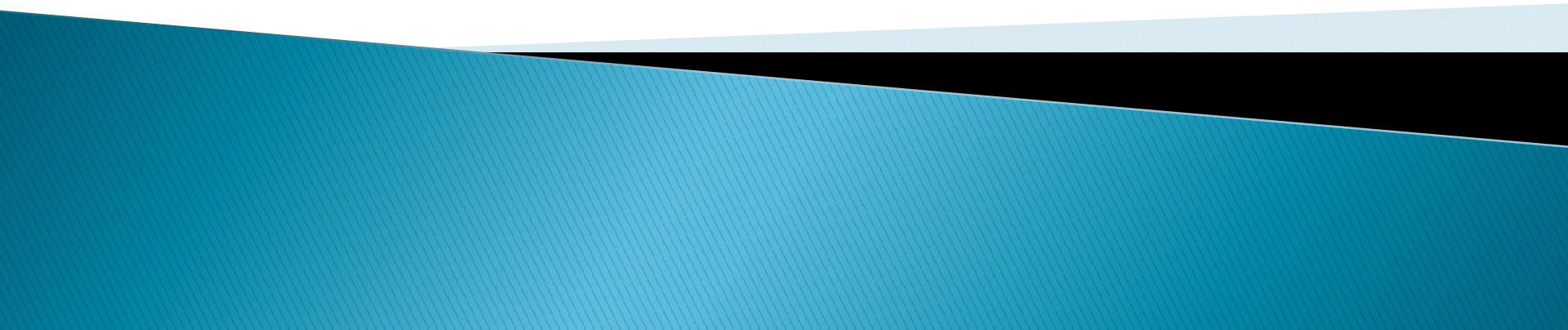


Computer Security and Cyber Law

Unit 6

Malicious Programs and Protection

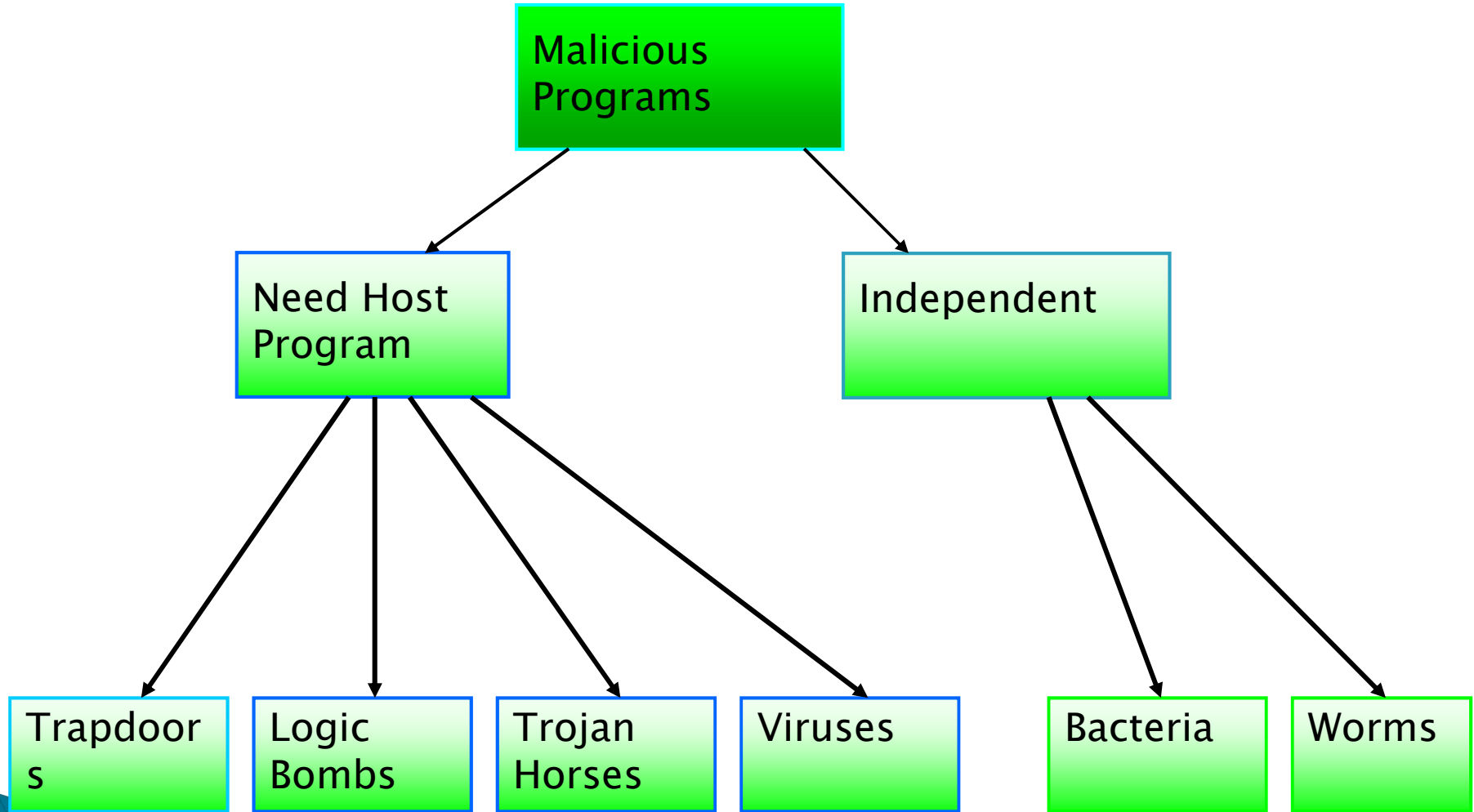
Lecturer: Binod Chandra Shrestha



Viruses and "Malicious Programs"

- ▶ Computer "Viruses" and related programs have the ability to replicate themselves on an ever increasing number of computers. They originally spread by people sharing floppy disks. Now they spread primarily over the Internet (a "Worm").
- ▶ Other "Malicious Programs" may be installed by hand on a single machine. They may also be built into widely distributed commercial software packages. These are very hard to detect before the payload activates (Trojan Horses, Trap Doors, and Logic Bombs).

Taxonomy of Malicious Programs



Definitions

- ▶ Virus – code that copies itself into other programs.
- ▶ A “Bacteria” replicates until it fills all disk space, or CPU cycles.
- ▶ Payload – harmful things the malicious program does, after it has had time to spread.
- ▶ Worm – a program that replicates itself across the network (usually riding on email messages or attached documents (e.g., macro viruses)).

Definitions

- ▶ Trojan Horse – instructions in an otherwise good program that cause bad things to happen (sending your data or password to an attacker over the net).
- ▶ Logic Bomb – malicious code that activates on an event (e.g., date).
- ▶ Trap Door (or Back Door) – undocumented entry point written into code for debugging that can allow unwanted users.
- ▶ Easter Egg – extraneous code that does something “cool.” A way for programmers to show that they control the product.

Definition

▶ Rabbits and Bacteria

- A Bacterium or a rabbit is a program that absorbs all of some class of resource.
- This malicious software replicates itself without limits. Depletes some or all the system's resources.
- Creates DOS attacks.
- Re-attacks the infected systems – difficult recovery.
- Exhausts all the system's resources such as CPU time, memory, disk space.
- Depletion of resources thus denying user access to those resources.

Defenses

- Defending against malicious logic takes advantage of several different characteristics of malicious logic to detect or to block, its execution.
- Sandboxing
 - Sandboxing are Virtual machines implicitly restrict process right.
 - Common implementation of this approach is to restrict the program by modifying it.
 - Special instructions inserted into the object code cause traps whenever an instruction violates the security policy.
 - The executable dynamically loads libraries, special libraries with the desired restrictions replace the standard libraries.

Defenses

▶ Information flow metrics

- This approach is to limit the distance a virus can spread.
- Malicious code usurps authority of user
- Limit information flow between users
 - If A talks to B , B can no longer talk to C
- Limits spread of virus
- Problem: Tracking information flow

▶ Reducing the rights

- The user can reduce her/his associated protection domain when running a suspect program.
- It follows the principle of least privilege.

▶ Malicious logic altering files

- Mechanisms using *manipulation detection codes (or MDCs)* apply some function to a file to obtain a set of bits called the *signature block* and then protect that block. If, after recomputing the signature block, the result differs from the stored signature block, the file has changed, possibly as a result of malicious logic altering the file. This mechanism relies on selection of good cryptographic checksums

Defenses

- ▶ Proof-carrying code
 - Code includes proof of correctness
 - At execution, verify proof against code
 - *If code modified, proof will fail*
- ▶ Notion of trust
 - Trust the user to take explicit actions to limit their process' protection domain sufficiently
 - That is, enforce least privilege correctly
 - Trust mechanisms to describe programs' expected actions sufficiently for descriptions to be applied, and to handle commands without such descriptions properly
 - Trust specific programs and kernel
 - Problem: these are usually the first programs malicious logic attack

Introduction to Virtual Machine

- ▶ Virtual Machine(VM) structure
 - A VM runs on a virtual monitor.
 - Monitor virtualizes the resources of the underlying system and presents to each VM the illusion that it and it alone is using the hardware.
- ▶ Virtual machine monitor(VMM)
 - The VMM runs at the highest level of privilege.
 - Keeps track of the state of each VM just as an ordinary Operating System keeps track of the state of its processes.
 - When a privileged instruction is executed, the hardware cause a trap to the VMM.
 - The monitor services the interrupt and restores the state of the caller.

Virus Phases

- ▶ **Dormant phase** – the virus is idle
- ▶ **Propagation phase** – the virus places an identical copy of itself into other programs
- ▶ **Triggering phase** – the virus is activated to perform the function for which it was intended
- ▶ **Execution phase** – the function is performed

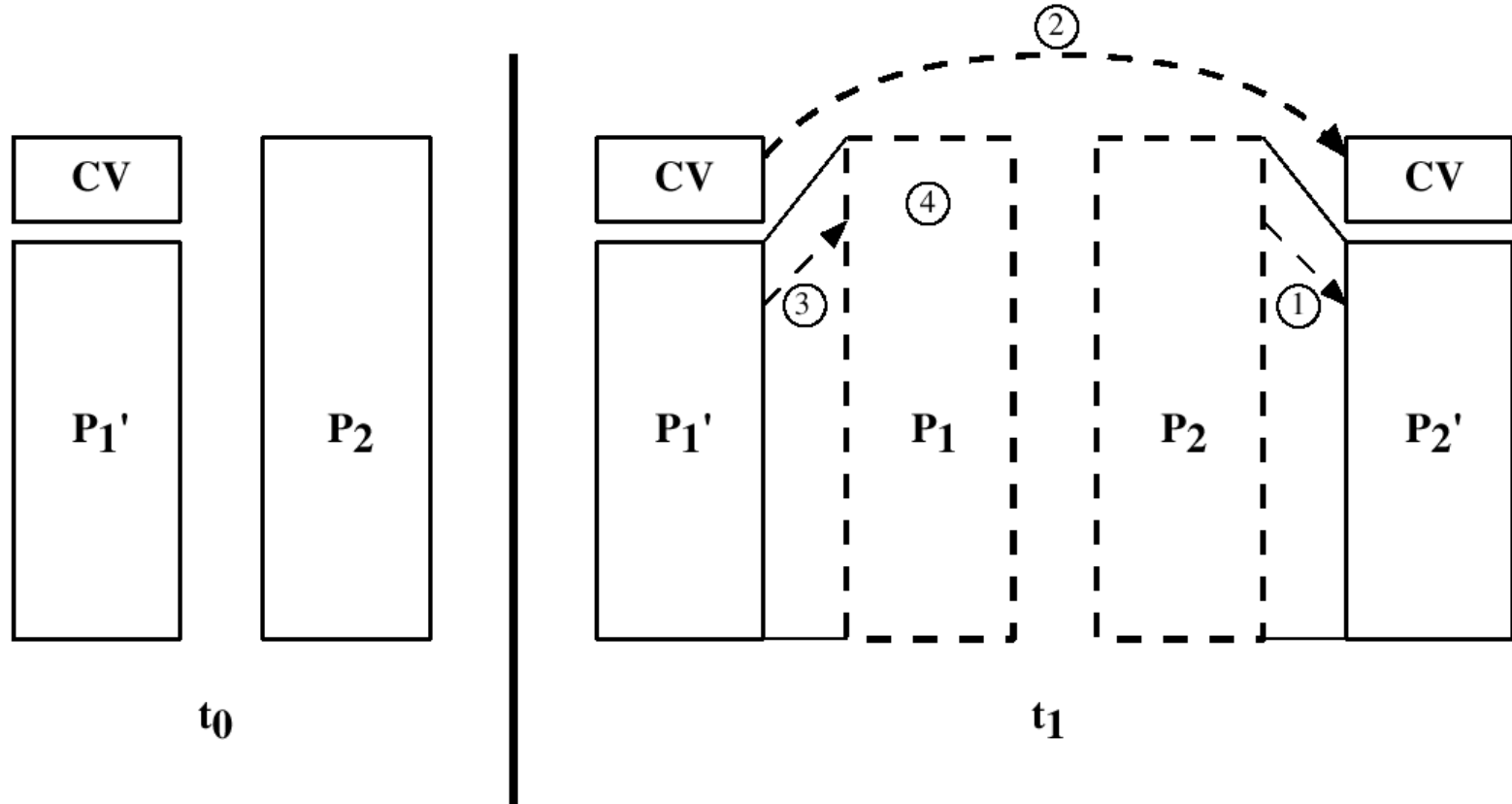
Virus Protection

- ▶ Have a well-known virus protection program, configured to
- ▶ scan disks and downloads automatically for known viruses.
- ▶ Do not execute programs (or "macro's") from unknown
- ▶ sources (e.g., PS files, Hypercard files, MS Office documents,
- ▶ Avoid the most common operating systems and email
- ▶ programs, if possible.

Virus Structure

```
program V :=  
  
  {goto main;  
   1234567;  
  
   subroutine infect-executable :=  
     {loop:  
      file := get-random-executable-file;  
      if (first-line-of-file = 1234567)  
        then goto loop  
        else prepend V to file; }  
  
   subroutine do-damage :=  
     {whatever damage is to be done}  
  
   subroutine trigger-pulled :=  
     {return true if some condition holds}  
  
main:  main-program :=  
      {infect-executable;  
       if trigger-pulled then do-damage;  
       goto next;}  
  
next:  
  
}
```

A Compression Virus



Types of Viruses

- ▶ **Parasitic Virus** – attaches itself to executable files as part of their code. Runs whenever the host program runs.
- ▶ **Memory-resident Virus** – Lodges in main memory as part of the residual operating system.
- ▶ **Boot Sector Virus** – infects the boot sector of a disk, and spreads when the operating system boots up (original DOS viruses).
- ▶ **Stealth Virus** – explicitly designed to hide from Virus Scanning programs.
- ▶ **Polymorphic Virus** – mutates with every new host to prevent signature detection.

Macro Viruses

- ▶ Microsoft Office applications allow “macros” to be part of the document. The macro could run whenever the document is opened, or when a certain command is selected (Save File).
- ▶ Platform independent.
- ▶ Infect documents, delete files, generate email and edit letters.

Antivirus Approaches

1st Generation, Scanners: searched files for any of a library of known virus “signatures.” Checked executable files for length changes.

2nd Generation, Heuristic Scanners: looks for more general signs than specific signatures (code segments common to many viruses). Checked files for checksum or hash changes.

3rd Generation, Activity Traps: stay resident in memory and look for certain patterns of software behavior (e.g., scanning files).

4th Generation, Full Featured: combine the best of the techniques above.

Advanced Antivirus Techniques

- ▶ Generic Decryption (GD)
 - CPU Emulator
 - Virus Signature Scanner
 - Emulation Control Module
- ▶ For how long should a GD scanner run each interpretation?

Advanced Antivirus Techniques

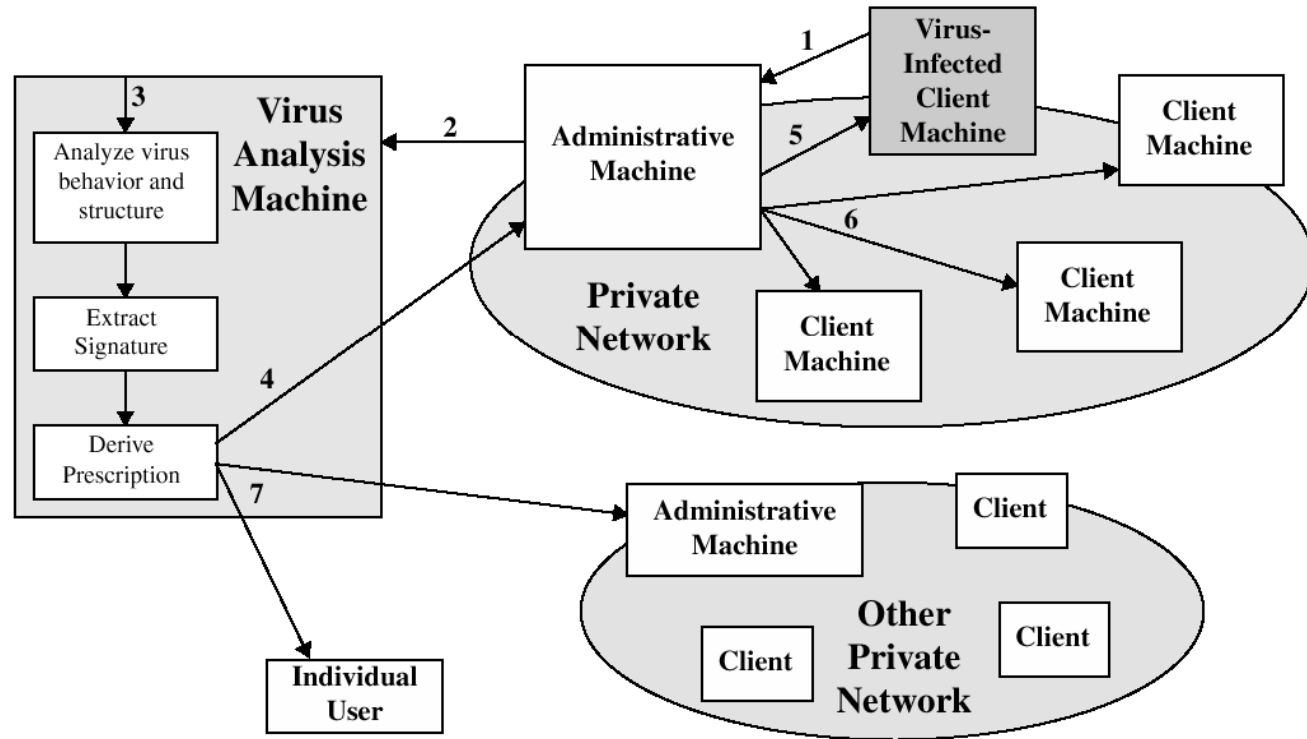


Figure 9.11 Digital Immune System

Assignment 6

- ▶ Explain taxonomy of malicious programs.
- ▶ What are differences between Virus and Trojan Horse?
- ▶ Explain simple structure of Virus program.
- ▶ What do you mean by be beneficial computer virus? Explain with suitable example.
- ▶ Describe about Advanced Antivirus Technique.